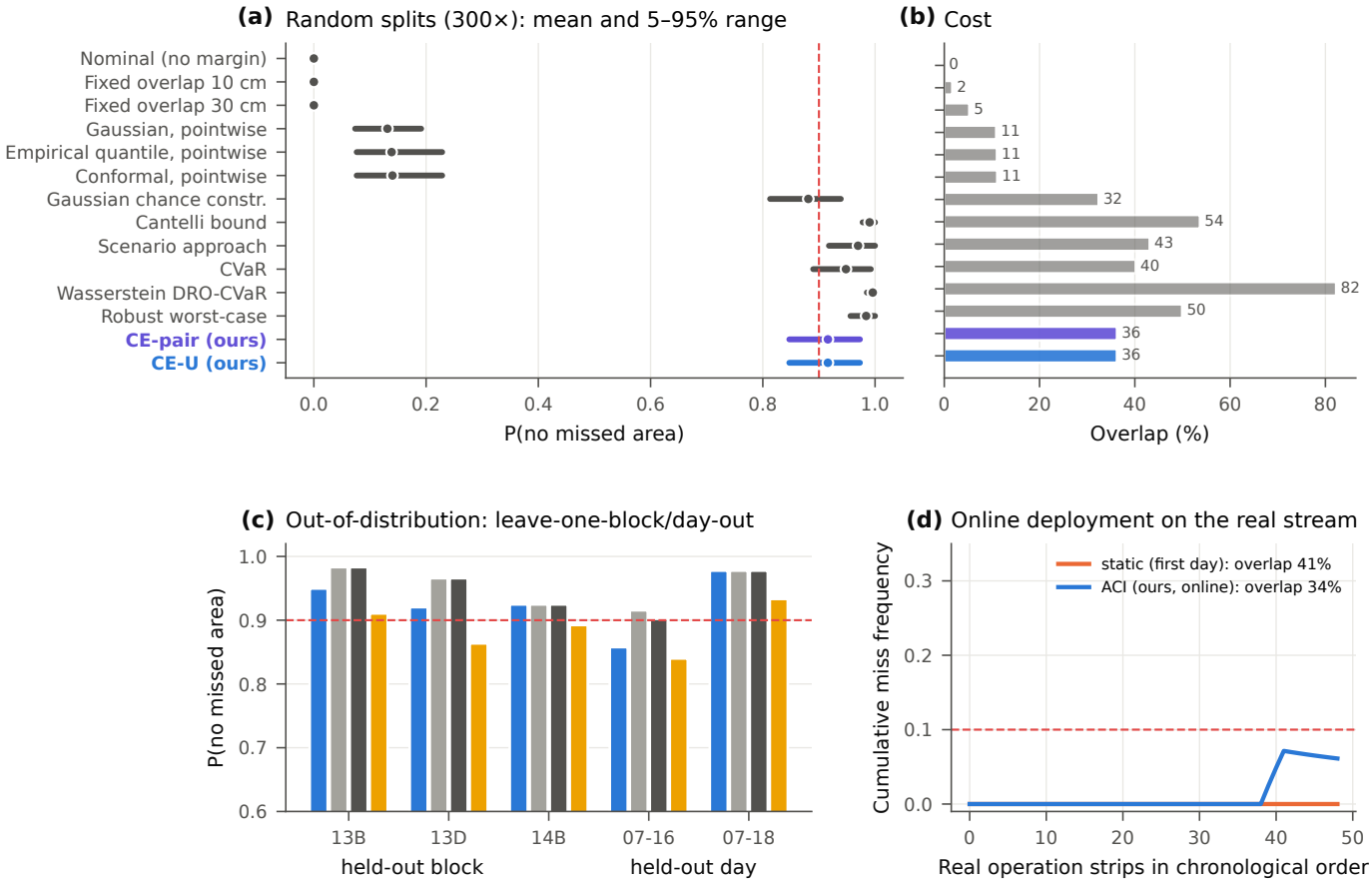
(a) Random splits (300×): mean and 5–95% range
Nominal (no margin)
Fixed overlap 10 cm
Fixed overlap 30 cm
Gaussian, pointwise
Empirical quantile, pointwise
Conformal, pointwise
Gaussian chance constr.
Cantelli bound
Scenario approach
CVaR
Wasserstein DRO-CVaR
Robust worst-case
CE-pair (ours)
CE-U (ours)
0.0
0.2
0.4
0.6
0.8
1.0
P(no missed area)
(b) Cost
0
2
5
11
11
11
32
54
43
40
82
50
36
36
0
20
40
60
80
Overlap (%)
(c) Out-of-distribution: leave-one-block/day-out
P(no missed area)
1.0
0.9
0.8
0.7
0.6
13B
13D
14B
07-16
07-18
held-out block
held-out day
(d) Online deployment on the real stream
static (first day): overlap 41%
ACI (ours, online): overlap 34%
Cumulative miss frequency
0.3
0.2
0.1
0.0
0
10
20
30
40
50
Real operation strips in chronological order
CE-U (ours)
Scenario approach
CVaR
Gaussian chance constr.